Forecasting the 2026 NCAA Tournament

A Simulation-Based Framework for Bracket Prediction and Cinderella Identification

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Abstract

This paper presents a multi-model framework for forecasting the 2026 NCAA Men's Basketball Tournament. I combine three distinct signals — a logistic regression win probability model trained on historical tournament matchups across all six rounds spanning 2011–2025 (excluding the canceled 2020 tournament), a Chebyshev distance-weighted nearest-neighbor classifier that maps each team to historically similar tournament performers, and a 10,000-iteration Monte Carlo bracket simulation — into a composite probability system that is grounded in team efficiency and sensitive to bracket path structure. The framework produces round-by-round survival probabilities, identifies likely deep runs and title contenders, and flags teams whose projected trajectories deviate meaningfully from the expectations implied by their seeding. Michigan emerges as the clear favorite at 23.9% composite championship probability, nearly double that of second-ranked Florida, with the result robust across all model components and bracket output methods. I additionally define the "Cinderella" classification for seeds 7–16 and produce probability estimates for each such team; the composite is anchored primarily by the Monte Carlo simulation (75%), with a Round of 64 matchup-specific upset model providing a 25% first-round context adjustment. Three bracket output methods: deterministic, modal, and upset threshold, are implemented to translate the probability estimates into actual brackets, each reflecting a different philosophy about how to handle the structural uncertainty that makes March Madness what it is.[1]

[1] I would like to thank Ken Pomeroy via Chris Toukmaji and Bart Torvik for their publicly available datasets that were used throughout my analysis.

Contents

1 Introduction **4**

2 Data and Features **5**

2.1 Data 5

2.2 Covariates 5

2.3 Outcome Variables 5

2.4 Bracket Structure and Data Ordering 6

3 Matchup Win Probability Model **6**

3.1 Training Data: All Tournament Rounds 6

3.2 Feature Construction 6

3.3 Logistic Regression Specification 7

4 Deep-Run Logistic Models **7**

4.1 Final Four and Championship Models 7

5 Chebyshev Distance-Weighted Nearest Neighbors **8**

5.1 Z-Score Normalization 8

5.2 Chebyshev Distance 8

5.3 Inverse-Distance Weighted Probability 9

6 Monte Carlo Bracket Simulation **9**

6.1 Simulation Algorithm 9

6.2 Output Metrics 10

7 Composite Probability Scores **10**

8 Defining and Measuring Cinderella Teams **11**

8.1 Definition 11

8.2 Cinderella Upset Model 12

8.3 Standalone Cinderella Quality Model 12

8.4 Composite Cinderella Probability 12

9 Bracket Output Methods **13**

9.1 Deterministic Most-Likely Bracket . . . 13

9.2 Upset Threshold Bracket . . . 13

9.3 Modal Bracket from Simulation . . . 14

10 Results and Bracket Projections **14**

10.1 Seed vs. Projected Wins . . . 14

10.2 Title Contender Tier Classification . . . 16

10.3 National Championship Probabilities . . . 18

10.4 Round-by-Round Survival . . . 21

10.5 Round by Round Probabilities of Winning each Round . . . 22

10.6 Cinderella Analysis Results . . . 24

10.6.1 Composite Cinderella Probabilities . . . 24

10.6.2 Round-by-Round Cinderella Survival . . . 26

10.6.3 Quality Versus Matchup Context . . . 27

10.7 Bracket Projections . . . 29

10.7.1 Deterministic Bracket . . . 29

10.7.2 Modal Bracket . . . 31

10.7.3 Upset Threshold Bracket . . . 32

10.7.4 My Bracket . . . 34

11 Limitations and Injury Adjustments **37**

12 Conclusion **38**

A Appendix: Historical Profiles by Round Reached **39**

1 Introduction

It is that time of year. I can hear the music already. As Jon Rothstein might say, grab your nitroglycerin pills — because with the coming of the third week of March, Americans across the country have grabbed their writing implement of choice in hopes of creating the first ever perfect bracket. This analysis hopes to modestly inform those decisions. The NCAA Division I Men's Basketball Tournament presents one of the most widely contested bracket prediction problems in sports analytics. Sixty teams (plus eight play-in teams) compete in a single-elimination format, producing a combinatorial space of roughly 9.2×10^{18} possible bracket outcomes. Despite this complexity, historical patterns in team quality metrics provide a meaningful signal for forecasting both individual game outcomes and deeper tournament runs. Furthermore, despite the advent of college basketball analytics from those such as Ken Pomeroy, Bart Torvik, and Evan Miyakawa among others, the perfect bracket has yet to be penciled in.

What makes March Madness so analytically compelling and as such, so persistently humbling is this tension between signal and noise. The tournament is long enough that the best teams win more often than they lose, and short enough that a single bad shooting night, a timely injury, or a fortunate bounce can unravel even the most carefully constructed bracket. The historical record bears this out: one-seeds win reach the second weekend an overwhelming proportion of the time but national champions have also come from lower seed lines, and double-digit seeds reach the Sweet Sixteen with enough regularity that treating upsets as anomalies is itself a modeling error. Thus, we must grapple with both halves of this reality simultaneously.

The proliferation of possession-based efficiency metrics over the past two decades has meaningfully improved the raw material available for bracket forecasting. Ken Pomeroy's adjusted efficiency ratings, Bart Torvik's BARTHAG, and related measures have given analysts tools to evaluate teams in ways that are substantially more predictive than raw win-loss records or scoring margins. These metrics adjust for opponent quality and pace of play, yielding estimates of how good a team actually is rather than simply how well it scheduled in the non-conference slate and has been scheduled in the conference slate. That said, translating these metrics into tournament probabilities is non-trivial. Teams that look identical in aggregate efficiency can have very different profiles. For example, one could be built on offensive firepower, another anchored by elite defense, and a third dependent on a pace that disrupts opponents.

This paper documents the full model used to generate 2026 NCAA Tournament bracket forecasts. The goals are threefold: to estimate the probability that the higher-seeded (or metrically stronger) team wins for any two teams that could meet in a given round; to identify which teams are most likely to reach the Sweet Sixteen, Elite Eight, Final Four, and ultimately win the national championship; and to identify those teams whose combination of team quality and favorable matchup context gives them an elevated probability of a memorable tournament run among the lower seeds (seeds 7–16), i.e., to identify potential Cinderellas.

The remainder of the paper proceeds as follows. Section 2 describes the data and feature construction, covering the KenPom and Barttorvik sources, the outcome encoding, and the bracket-structured ordering that is preserved throughout the analysis. Section 3 develops the matchup-level win probability model. Section 4 introduces the standalone deep-run logistic models, which predict Final Four and championship outcomes from team-level metrics alone, independent of bracket draw. Section 5 describes the Chebyshev distance-weighted nearest-neighbor classifier. Section 6 details the Monte Carlo bracket simulation and

Section 7 explains how the Monte Carlo output and standalone quality models are combined into composite championship and Final Four probability estimates. Section 8 defines the Cinderella framework, including its seed-based definition, a first-round upset model, and the composite Cinderella probability. Thereafter I identify the 2026 field's best deep-run candidates from the lower seed lines. Section 9 describes the three bracket output methods. Section 10 presents the full results, including seed-versus-projected-wins comparisons, title contender tier classifications, round-by-round survival curves, Cinderella analysis, and four bracket projections including my own. Section 11 closes with a discussion of the framework's limitations and Section 12 concludes.

2 Data and Features

2.1 Data

All data used in this analysis come from two widely used college basketball analytics platforms: Ken Pomeroy's `kenpom.com` and Bart Torvik's `barttorvik.com`. These sources provide advanced statistics designed to evaluate team performance using possession-based efficiency metrics and schedule-adjusted ratings. The datasets from these two sites were merged to construct the modeling dataset used throughout the analysis. The dataset covers NCAA Tournament participants from 2011 through 2025, comprising 896 team-seasons after excluding the canceled 2020 tournament. Each observation includes adjusted offensive and defensive efficiency metrics from KenPom and Barttorvik, strength-of-schedule measures, quality win indices, and round-by-round outcome indicators among other team-level measurements.

2.2 Covariates

The models use a set of variables intended to capture overall team strength and performance across several dimensions commonly emphasized in modern college basketball analytics. The feature set spans measures of offensive and defensive efficiency and multiple indicators of strength of schedule, and includes additional variables capturing aspects of team performance such as results against stronger competition and deviations from expected outcomes.

2.3 Outcome Variables

Each team-season carries binary win indicators for each round: `W_64` (won the Round of 64), `W_32` (won the Round of 32 game), through `W_2` (won the national championship game). Total wins are summed into `wins_total` $\in \{0, 1, 2, 3, 4, 5, 6\}$, where 6 denotes a national champion.

Note that First Four teams play their opening game as a Round of 64 contest in the data encoding. A First Four team with `wins_total` $= 6$ therefore won six games including their opening game and is a national champion — they are not a team that won the First Four game and then lost in the championship. This is an important distinction for correctly interpreting win thresholds.

2.4 Bracket Structure and Data Ordering

A critical design feature of this dataset is that row ordering encodes bracket structure. Teams are stored in consecutive pairs: positions 1–2 meet in the first round, positions 3–4 meet in the first round, and so forth. Within each region of 16 teams, the winner of game 1 (positions 1–2) plays the winner of game 2 (positions 3–4) in the second round. This structure is strictly preserved throughout all data manipulation and simulation steps.

3 Matchup Win Probability Model

3.1 Training Data: All Tournament Rounds

A key methodological choice in this framework is that the win probability model is trained on matchups from all six tournament rounds, not only the Round of 64. The motivation is straightforward: a model trained exclusively on a smaller subset of games where seed gaps and metric differentials will most likely be larger by design will have a harder time predicting the later rounds where both teams have already survived at least one game and metric gaps begin to narrow. Pooling across all rounds yields a richer and more representative training set, and crucially, a model that is better calibrated for the close, high-leverage games that define the back half of the bracket.

Matchups for each round are extracted from the bracket-structured data using a pod-based approach:

Round	Pod size	Qualifier	Matchups
R64	2	(none)	448
R32	4	`W_64 = 1`	224
S16	8	`W_32 = 1`	111
E8	16	`W_16 = 1`	56
FF	32	`W_8 = 1`	28
CG	64	`W_4 = 1`	14
Total			**881**

For each pod, the two qualifying teams are identified, the higher-seeded team is designated the favorite, and metric differentials are computed. This yields 881 matchup observations spanning all rounds of tournament play from 2011 through 2025.

3.2 Feature Construction

Rather than modeling teams in isolation, the win probability model is built around matchup differentials, i.e., the gap between the two competing teams on each metric. This framing reflects the fundamentally relative nature of tournament outcomes: what matters is not how good a team is in absolute terms, but

how it compares to its projected opponents on Selection Sunday. For each matchup, the differential of each metric is computed between the underdog (lower seed) and the favorite (higher seed). Formally, for a matchup between favorite f and underdog d:

$$\Delta \mathbf{x} = \mathbf{x}^{(d)} - \mathbf{x}^{(f)} \tag{1}$$

This differential framing captures the relative gap between teams rather than their absolute quality. A positive entry $\Delta x_k > 0$ indicates the underdog outperforms the favorite on dimension k — a structural indicator of potential upset. The outcome variable $Y = 1$ if the favorite wins, and 0 otherwise.

3.3 Logistic Regression Specification

With binary win/loss outcomes and a clean set of continuous differentials as inputs, logistic regression is a natural and interpretable choice for the win probability model. It produces well-calibrated probabilities and keeps the model interpretable enough to interrogate which dimensions of team quality are most predictive. The win probability model is a logistic regression estimated on the full set of 881 cross-round matchup observations:

$$\log\left(\frac{P(\text{fav wins})}{1 - P(\text{fav wins})}\right) = \beta_0 + \boldsymbol{\beta}^\top \Delta \mathbf{x} + \varepsilon \tag{2}$$

where $\Delta \mathbf{x} \in \mathbb{R}^p$ is the vector of matchup differentials and $\boldsymbol{\beta}$ is the corresponding coefficient vector estimated via maximum likelihood. Because differentials are computed as underdog minus favorite, entries of $\Delta \mathbf{x}$ are typically negative when the favorite is better, and positive $\hat{\beta}_k$ coefficients indicate association with upset likelihood.

4 Deep-Run Logistic Models

The matchup model is well-suited for head-to-head win probabilities, but it is inherently path-dependent. That is to say, a team's simulated championship probability reflects not just its quality, but the luck of its draw. To complement this, standalone team-quality models are estimated that predict deep tournament runs from team-level metrics alone, independent of draw or bracket path. The idea is to ask the qualifying question: does this team look like a Final Four team or a champion, based purely on its profile? These models serve as a quality floor that anchors the final composite estimates.

4.1 Final Four and Championship Models

The Final Four model takes the form:

$$\log\left(\frac{P(\text{Final Four})}{1 - P(\text{Final Four})}\right) = \alpha_0 + \boldsymbol{\alpha}^\top \mathbf{x} + \varepsilon \tag{3}$$

where $\mathbf{x} \in \mathbb{R}^q$ is the team-level feature vector and $\boldsymbol{\alpha}$ is estimated via maximum likelihood on all 896 historical tournament teams from 2011–2025. An identical specification is estimated for $P(\text{Champion})$, replacing the outcome with $\mathbf{1}[\texttt{wins_total} = 6]$. An Elite Eight model ($\mathbf{1}[\texttt{wins_total} \geq 3]$) is also estimated for intermediate deep-run scoring. Thus, the standalone model captures whether a team "looks like" a historical champion irrespective of draw. In this vein, historical champions from 2011–2025 averaged an adjusted offensive efficiency of 122.6, an adjusted defensive efficiency of 91.8, a wins above bubble of 9.0, and a KenPom rank of 5.6.

5 Chebyshev Distance-Weighted Nearest Neighbors

Logistic regression imposes a linear decision boundary, which works well on average but can miss more complex patterns. For instance, the possibility that a team must clear multiple thresholds simultaneously to be a viable Final Four contender. A team that is elite offensively but lackluster defensively may look acceptable in a linear model while being systematically exploitable by good coaching staffs in a six-game stretch. To capture these kinds of joint threshold effects, a Chebyshev distance-weighted nearest-neighbor classifier is employed alongside the logistic model. Rather than fitting a global parametric surface, this approach attempts to understand which historical tournament teams a given team most closely resembles and uses their outcomes as a weighted vote.

5.1 Z-Score Normalization

Before computing distances, all covariates are standardized so that no single metric dominates by virtue of its scale:

$$z_k^{(t)} = \frac{x_k^{(t)} - \mu_k}{\sigma_k}$$

where μ_k and σ_k are computed on all historical tournament teams for the deep-run models, and on historical tournament teams seeded 7 or lower for the Cinderella-specific model, ensuring that distance calculations reflect within-tier comparisons in the latter case.

5.2 Chebyshev Distance

The choice of distance metric is deliberate. Rather than Euclidean distance — which averages across all dimensions and can allow a single catastrophic weakness to be masked by strengths elsewhere — the Chebyshev (supremum) norm is used. The distance between team t and historical team s is:

$$d_\infty(t, s) = \max_{k \in \{1,\ldots,p\}} \left| z_k^{(t)} - z_k^{(s)} \right| \tag{4}$$

This metric ensures that a team which is markedly deficient on any single dimension — for instance, a

below average offensive efficiency — is penalized regardless of how strong it may be on other dimensions. This is appropriate since a single glaring weakness can be systematically exploited by good coaches.

5.3 Inverse-Distance Weighted Probability

Historical comparables are then aggregated using inverse-distance weighting, so that teams more similar to the query team exert greater influence on the probability estimate. The probability that team t achieves at least τ wins is thus:

$$\hat{P}(\texttt{wins_total} \geq \tau \mid t) = \frac{\sum_{s=1}^{N} w_s \cdot \mathbf{1}[\texttt{wins_total}^{(s)} \geq \tau]}{\sum_{s=1}^{N} w_s}, \qquad w_s = \frac{1}{d_\infty(t,s) + \epsilon} \tag{5}$$

where $\epsilon = 0.01$ prevents division by zero when a 2026 team is an exact match to a historical team in normalized feature space.

6 Monte Carlo Bracket Simulation

6.1 Simulation Algorithm

The logistic and nearest-neighbor models both produce team-level probability estimates, but neither accounts for the bracket path a team must actually navigate. A #1 seed in a brutal region faces a different championship landscape than an identically-rated team on the other side of the draw. To capture this path dependency, the win probabilities from the matchup model are fed into a Monte Carlo bracket simulation. Running 10,000 simulated tournaments allows the full distribution of bracket outcomes to emerge naturally from the model, rather than having to be imposed analytically.

For each of the 10,000 Monte Carlo iterations, the tournament field is initialized with teams indexed $1, 2, \ldots, 64$ in bracket-pair order. In each round, the algorithm iterates through consecutive team pairs (i, j) in the current field and simulates the game outcome by drawing $u \sim \text{Uniform}(0, 1)$. If $u < W_{ij}$, team i advances; otherwise team j advances, where $W_{ij} = P(\text{team } i \text{ beats team } j)$. This process continues round by round until a single champion remains, and the total number of wins accumulated by every team is recorded at the conclusion of each simulation.

The bracket path is strictly preserved throughout: the winner of game $(2k - 1)$ plays the winner of game $2k$ in the next round, not the winner of game $(2k + 1)$. This correctly captures the bracket structure as a #1 seed cannot face another #1 seed before the Final Four. The random seed is fixed at 42.

6.2 Output Metrics

From the wins matrix $\mathbf{V} \in \mathbb{Z}^{64\times 10000}$ (where $V_{t,s}$ is the number of wins team t accumulated in simulation s), the probability that team t won each respective round's game is:

$$\hat{P}_t(\text{won R64 game}) = \frac{1}{N}\sum_{s=1}^{N} \mathbf{1}[V_{t,s} \geq 1] \tag{6}$$

$$\hat{P}_t(\text{won R32 game}) = \frac{1}{N}\sum_{s=1}^{N} \mathbf{1}[V_{t,s} \geq 2] \tag{7}$$

$$\hat{P}_t(\text{won S16 game}) = \frac{1}{N}\sum_{s=1}^{N} \mathbf{1}[V_{t,s} \geq 3] \tag{8}$$

$$\hat{P}_t(\text{won E8 game}) = \frac{1}{N}\sum_{s=1}^{N} \mathbf{1}[V_{t,s} \geq 4] \tag{9}$$

$$\hat{P}_t(\text{won FF game}) = \frac{1}{N}\sum_{s=1}^{N} \mathbf{1}[V_{t,s} \geq 5] \tag{10}$$

$$\hat{P}_t(\text{Champion}) = \frac{1}{N}\sum_{s=1}^{N} \mathbf{1}[V_{t,s} \geq 6] \tag{11}$$

with $N = 10{,}000$. Note the interpretation: $\hat{P}_t(\text{won R64 game})$ is the probability of winning the Round of 64 game, equivalently the probability of advancing to the Round of 32; $\hat{P}_t(\text{won R32 game})$ is the probability of winning the Round of 32 game, equivalently advancing to the Sweet Sixteen, and so on.

7 Composite Probability Scores

Each component model captures something real but incomplete on its own. The Monte Carlo simulation is path-aware but can be sensitive to draw luck and the standalone logistic and Chebyshev models reflect team quality independent of bracket position but are blind to who a team actually has to go through. Combining them produces estimates that are grounded in team quality and sensitive to bracket realities. The final championship and Final Four probability estimates are therefore composites of the path-aware Monte Carlo output and the standalone quality models:

$$\hat{P}_t^{\text{standalone}}(\text{FF}) = \tfrac{1}{2}\,\hat{P}_t^{\text{Cheb}}(\text{FF}) + \tfrac{1}{2}\,\hat{P}_t^{\text{logit}}(\text{FF}) \tag{12}$$

$$\hat{P}_t^{\text{standalone}}(\text{Champ}) = \tfrac{1}{2}\,\hat{P}_t^{\text{Cheb}}(\text{Champ}) + \tfrac{1}{2}\,\hat{P}_t^{\text{logit}}(\text{Champ}) \tag{13}$$

$$\hat{P}_t^{\text{composite}}(\text{FF}) = 0.60 \cdot \hat{P}_t^{\text{MC}}(\text{won E8}) + 0.40 \cdot \hat{P}_t^{\text{standalone}}(\text{FF}) \tag{14}$$

$$\hat{P}_t^{\text{composite}}(\text{Champ}) = 0.60 \cdot \hat{P}_t^{\text{MC}}(\text{Champ}) + 0.40 \cdot \hat{P}_t^{\text{standalone}}(\text{Champ}) \tag{15}$$

The composite championship probabilities are subsequently normalized to sum to one across all 64 teams, ensuring a valid probability distribution over the full field. The 60/40 weighting toward Monte Carlo reflects the view that bracket path difficulty is the dominant factor in tournament outcomes as two identically-rated teams will have materially different championship probabilities depending on the strength of their regional draw which could unequally distribute factors such as accumulated mental and physical fatigue. The 40% standalone component provides a quality floor that prevents a strong team from being excessively penalized for a single difficult early matchup.

8 Defining and Measuring Cinderella Teams

8.1 Definition

Part of what makes March Madness compelling is the Cinderella story — the higher seeded team that keeps winning when it has no business doing so. Quantifying that phenomenon requires a precise operational definition, since "upset" means something different depending on the seed. A "Cinderella" team is defined as:

$$\text{Cinderella}(t) = \begin{cases} 1 & \text{if } \texttt{seed}(t) \in \{7, 8, 9, 10\} \text{ and } \texttt{wins_total}(t) \geq 3 \\ 1 & \text{if } \texttt{seed}(t) \geq 11 \text{ and } \texttt{wins_total}(t) \geq 2 \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

Seeds 7–10 must win their Sweet Sixteen game (i.e. reach the Elite Eight) to qualify; seeds 11 and above must win their Round of 32 game (i.e. reach the Sweet Sixteen). This asymmetry reflects the fact that a 7-seed defeating a 10-seed in the Round of 64 is the expected result and there is a world in which said 7-seed is able to knock off the 2-seed in their region followed by the winner of the 3-6 game above them, the world in which an 11-seed reaches the Sweet Sixteen typically requires defeating a 6-seed followed by a 3-seed, two substantially better metrically teams. In my eyes, said latter world is less likely and more, well, Cinderella-esque also given the fact that teams at the seven line are typically power conference teams while those at the 11 line have a higher frequency of being mid-majors. Thus, the definition really counts how many games a Cinderella is not favored to win and sets the threshold at two. While ten seeds are usually metrically disadvantaged against the typical 7-seed, 10 seeds win at a historical clip substantial enough

for me to group them along with the 7 through 9 seeds in the same stratum. Should I choose to alter my definition and group them with the 11 or lower seeds, my model would certainly be higher on the 10 seeds as a group.

8.2 Cinderella Upset Model

Predicting Cinderella runs requires a model tuned specifically to the first-round context in which they begin. The general win probability model is trained across all six rounds and all seed ranges, which makes it well-calibrated on average but potentially less sensitive to the specific dynamics of higher-seed matchups in the Round of 64. To address this, a separate logistic regression is estimated exclusively on the 224 historical Round of 64 matchups involving seeds 7 and below:

$$\log\left(\frac{P(\text{upset})}{1 - P(\text{upset})}\right) = \gamma_0 + \boldsymbol{\gamma}^\top \Delta\mathbf{x}^{\text{cin}} + \varepsilon \tag{17}$$

where $\Delta\mathbf{x}^{\text{cin}} \in \mathbb{R}^r$ is a Cinderella-specific differential feature vector that includes variables absent from the general win probability model — most notably measures of game luck and quality game exposure — that have demonstrated particular predictive value in the Round of 64 context for higher-seeded teams. That is, exposure to quality opponents in the non-conference schedule helps with experience and it is always nice to be more lucky than your opponents. This model is deliberately restricted to Round of 64 matchups; Cinderella classification is inherently first-round-context specific and extending it to later rounds would dilute its calibration for the setting in which it is most useful.

8.3 Standalone Cinderella Quality Model

Beyond the opening game, a team's broader profile also matters — some lower seeds simply look more like historical Cinderellas than others, regardless of their first-round opponent. The standalone Cinderella quality signal captures this by combining the Chebyshev classifier and a Cinderella-specific logistic regression, both normalized and estimated on the subset of historical tournament teams seeded 7 or lower:

$$\hat{P}_t^{\text{standalone,cin}} = \tfrac{1}{2}\,\hat{P}_t^{\text{Cheb,cin}} + \tfrac{1}{2}\,\hat{P}_t^{\text{logit,cin}} \tag{18}$$

The logistic component uses a stepwise-confirmed optimal feature set estimated on the same historical pool. The Chebyshev component uses normalization statistics derived from that same pool, ensuring that distance calculations reflect within-tier comparisons.

8.4 Composite Cinderella Probability

The composite Cinderella probability brings together the first-round upset signal and the path-aware simulation, weighting the Monte Carlo output more heavily since a Cinderella run is by definition a multi-game achievement that the simulation captures directly:

$$\hat{P}_t^{\text{Cinderella}} = 0.25 \cdot \hat{P}_t^{\text{upset}} + 0.75 \cdot \hat{P}_t^{\text{MC,cin}} \tag{19}$$

where $\hat{P}_t^{\text{upset}}$ is the Round of 64 upset model probability and

$$\hat{P}_t^{\text{MC,cin}} = \begin{cases} \hat{P}_t(\text{won S16 game}) & \text{if } \texttt{seed}(t) \in \{7, \ldots, 10\} \\ \hat{P}_t(\text{won R32 game}) & \text{if } \texttt{seed}(t) \geq 11 \end{cases} \tag{20}$$

is the round-appropriate Monte Carlo Cinderella frequency. The standalone quality signal enters the composite indirectly but carries no direct weight in $\hat{P}_t^{\text{Cinderella}}$. The rationale is twofold: first, a Cinderella run requires winning at least two games, and a bracket-agnostic quality signal adds limited marginal value once a path-aware simulation is available; second, the Monte Carlo already incorporates team quality implicitly, as teams with stronger metrics receive higher win probabilities in every simulated game. The upset model retains 25% weight as a first-round context modifier, rewarding teams with a favorable opening draw beyond what the general simulation would otherwise capture.

9 Bracket Output Methods

Probability estimates are nice and useful, but a bracket competition requires discrete picks. Three methods of translating continuous win probabilities into a filled bracket are implemented, each reflecting a different philosophy about how to handle uncertainty.

9.1 Deterministic Most-Likely Bracket

The simplest bracket projection advances the team with $W_{ij} > 0.5$ in every game. While consistent, this approach is conservative: a bracket of 63 games in which every favorite wins would be incredibly atypical. Based on historical rates, a realistic 64-team bracket produces approximately 10–15 upsets.

9.2 Upset Threshold Bracket

To address the over-conservatism of the deterministic bracket, an upset threshold method is implemented. Rather than always picking the favorite, any game where the underdog has a non-trivial probability of winning is decided probabilistically. A parameter $p_{\min}$ defines the minimum underdog win probability required to trigger a probabilistic decision:

$$\text{winner} = \begin{cases} \text{favorite} & \text{if } P(\text{underdog wins}) < p_{\min} \\ \text{Bernoulli}(W_{ij}) & \text{if } P(\text{underdog wins}) \geq p_{\min} \end{cases} \tag{21}$$

The default setting $p_{\min} = 0.25$ means any game where the underdog has at least a 25% chance is decided

by a weighted coin flip using the actual model probabilities. Projected mismatches like a 1-seed versus 16-seed, for instance, where the underdog's probability is near 0% still go deterministically to the favorite.

9.3 Modal Bracket from Simulation

The most principled bracket output comes directly from the simulation itself. Rather than imposing a decision rule on top of the probabilities, this approach tracks which team occupied each bracket slot most frequently across all 10,000 simulations thereby letting the bracket be molded by the data rather than from an exogenous rule. For each slot g at round r, a slot count matrix records how many simulations placed each of the 64 teams in that position. The modal occupant is then selected:

$$\text{modal winner}(r, g) = \arg\max_{t \in \{1,\ldots,64\}} \text{SlotCount}_{r,g,t} \tag{22}$$

This approach naturally produces upsets wherever the simulation generated them frequently, respects path dependencies (a team cannot be the modal winner of a slot it rarely reached), and outputs two quantities per game: the head-to-head win probability W_{ij} and the simulation frequency for that slot.

10 Results and Bracket Projections

10.1 Seed vs. Projected Wins

Figure 1 is, in many respects, the most honest graph in this paper. It makes no pretense of championship probability or bracket architecture; it simply asks how many games each team is projected to win and then holds that answer up against fifteen years of what teams at that seed line have actually done. What we get is a portrait of a tournament field that is simultaneously top-heavy and unusually competitive in the middle, with several seed lines carrying teams that would be entirely at home two or three lines down.

Start at the top. Michigan sits noticeably above the historical one-seed expectation — not merely above the interquartile band but above the upper whisker of it. This is meaningful. One-seeds are not a uniform bloc; the best of them, the teams that cut down nets, consistently post projected wins in the 3.8–4.5 range, and Michigan's marker is pushing firmly toward the upper end of that corridor. Florida, also a one-seed, lands nearly as high. Duke, by contrast, sits closer to the historical mean for a one-seed — still a legitimate title contender, but carrying a path and an efficiency profile that the model reads as more conventional. Arizona, the fourth one-seed, falls below the other three.

The two-seed line tells an equally interesting story. Purdue is the outlier: its marker hovers conspicuously above the historical average, reinforcing the narrative that the Boilermakers are functionally underseeded. A team projected to win at the two-seed level what most teams win as a one-seed is a problem for every opponent in the West region. Houston, Iowa State, and UConn, the other two-seeds, track closer to expectations, though none dip materially below the mean.

The three-seed tier is where the field starts to reveal its teeth. Illinois and Gonzaga both project above

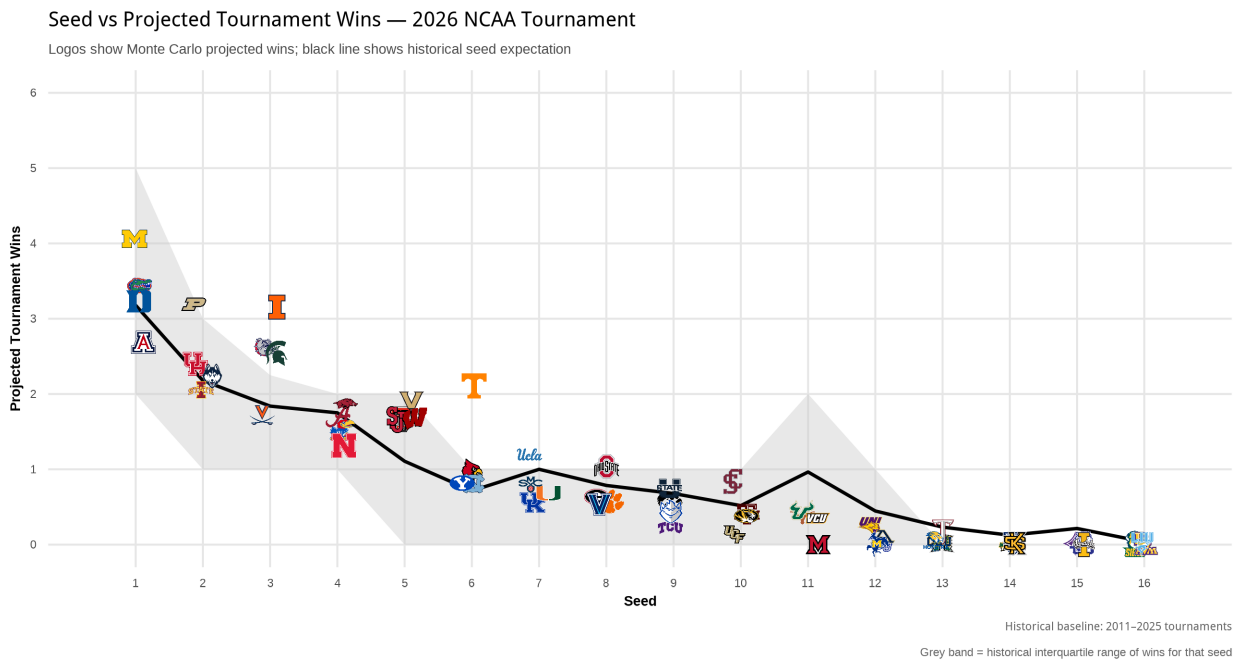

Figure 1: Projected tournament wins plotted against seed for the 2026 NCAA Tournament. Each logo represents a team's Monte Carlo projected win total. The solid black line denotes the historical seed-level mean from 2011–2025, and the shaded band captures the interquartile range of historical outcomes. Teams above the band represent meaningful overachievers relative to seed expectation; teams below it face structurally harder paths or carry weaker efficiency profiles than is typical for their seed line.

the historical three-seed mean by a substantial margin, and Michigan State — seeded third in the East — similarly outperforms its seed-line expectation. Taken together, this implies that the teams most likely to face these three are walking into stiffer tests than they might expect from a cursory glance at the bracket.

The five-seed line deserves particular attention. As a group, the fives in this field cluster above the historical average, which means the twelve-seeds opposing them will face a harder path than the historical 12-over-5 upset rate would suggest. Anyone building a bracket around the reliable 12-over-5 upset should note that this year's fives are not the usual assortment; they are, as a group, better than that and all project as four seeds. Tennessee, seeded sixth in the Midwest, is perhaps the most dramatic outlier at its seed line: its projected win total approaches what is typically associated with a strong four or even a three, making it a legitimate bracket-buster.

Further down the distribution, UCLA headlines a weaker 7-seed line and Ohio State sits above its seed-line expectations as an eights. Utah State on the 9 line headlines an average group of 9 seeds and Santa Clara on the ten line similarly exceeds the historical baseline, providing theoretical underpinning for its appearance later in the Cinderella analysis. The higher seeds — elevens through sixteens — largely compress against zero as expected, with a handful of modest exceptions. Northern Iowa, South Florida, Virginia Commonwealth, Texas, and Troy both sit just barely above zero but meaningfully above the historical floor, suggesting that their upset potential is at least partly grounded in underlying team quality rather than purely favorable matchup structure.

In regards to structure of the tournament, we should note the fact that historically, 11 seeds win more games than 10, 9, and 8 seeds, a bug of the tournament format. This is due to the fact that 11 seeds have winnable Round of 64 games against 6-seeds and then while harder matchups, still winnable Round of 32 games against 3-seeds while 8, 9, and 10 seeds have Round of 32 matchups with the stronger 1 and 2 seeds respectively.

10.2 Title Contender Tier Classification

The top 10 teams by $\hat{P}^{\text{composite}}(\text{Champ})$ are classified into tiers based on their offensive and defensive efficiency gaps relative to the historical champion archetype ($\bar{x}_{\texttt{adjoe}} = 122.6$, $\bar{x}_{\texttt{adjde}} = 91.8$):

Tier	P(Champ) Range	Profile
Elite — Complete Champion Profile	$\geq 18\%$	Exceeds archetype on both O & D
Contender — Defense Anchored	$\geq 10\%$	Defense gap $< +1.5$ pts
Contender — Offensive Powerhouse	$\geq 10\%$	Offense gap $> +3$ pts
Contender — Well-Rounded	$\geq 10\%$	Neither extreme
Fringe — Offense Below Champion Bar	$\geq 5\%$	Offense gap < -1 pt
Fringe — Path Dependent	$\geq 5\%$	Favorable draw required

Figure 2 gives us the tournament race as a geometry problem. The 2026 contenders are spread across the efficiency plane in ways that tell structurally distinct stories about what each team actually is and what it would need to do to cut down the nets.

The upper-right quadrant, representing teams that exceed the historical champion benchmark on both offense and defense, is where the truest title profiles live. Michigan occupies this space most definitively, combining an adjusted offensive efficiency meaningfully above the champion average with a defensive profile that is similarly elite. The logo size — scaled to composite championship probability — confirms what the coordinates imply: Michigan is not merely the best team in this quadrant but among the best-fitting championship profiles the model has encountered across the historical window. Duke and Arizona also reside here, and it is worth being precise about how they differ. Duke sits closer to the center of the quadrant, its two-way efficiency balance resembling the clean, well-rounded construction of teams like 2015 Duke or 2022 Kansas. In fact, Duke's "fit" is better than Michigan's. Arizona, by contrast, is pulled further left on the offensive axis, while its defensive numbers still clear the historical bar.

Florida and Houston occupy the boundary zone between the upper-right and lower-right quadrants — teams whose defense is near the champion average but whose offensive efficiency gap is narrower than the Michigan–Duke–Arizona cluster. In the lower right quadrant, Illinois and Purdue sit far to the right on the offensive axis and as such carry adjusted offensive efficiencies comfortably above the champion average but also perform meaningful below the historical average on the defensive end of the floor. These are teams that win games by putting points on the board faster than opponents can answer, and they do it at an elite level. Purdue is the more extreme case, its marker pushed furthest to the right of all teams in the figure; its offensive gap versus the champion archetype is substantial, and its defensive gap below the champion

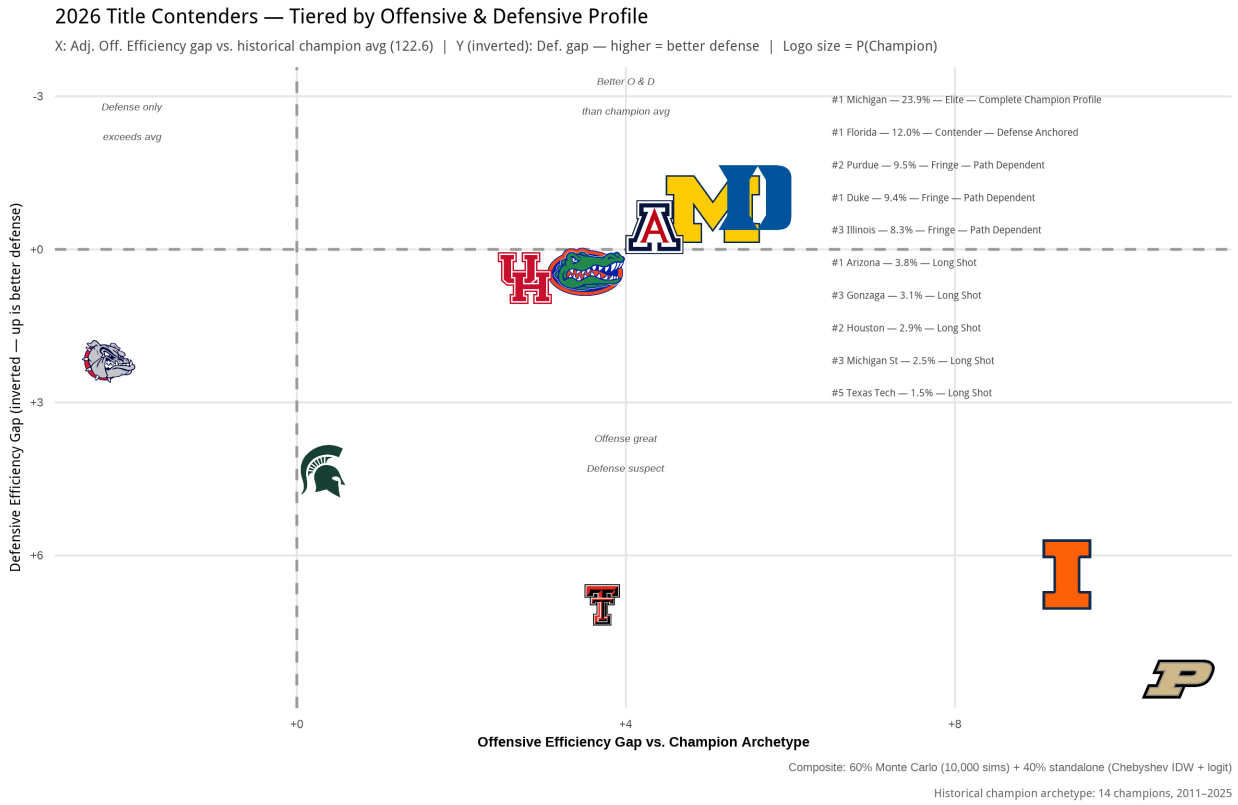

Figure 2: 2026 title contenders positioned by offensive and defensive efficiency gap relative to the historical national champion archetype (2011–2025). The horizontal axis measures adjusted offensive efficiency minus the historical champion mean; the vertical axis measures adjusted defensive efficiency gap, inverted so that higher positions indicate stronger defense. Dashed reference lines denote the champion-average benchmarks on each dimension. Logo size scales with composite championship probability. Top 10 teams by probability are displayed.

average is equally so. This profile carries championship upside. Indeed, offensive powerhouses have won titles, but it also carries a specific vulnerability: when the offensive margin diminishes against an elite opponent due to a poor shooting night or a stronger defense, there is no second wall to lean on. Illinois sits to Purdue's left but shares the structural profile. Texas Tech now without their best player JT Toppin who would have made them a serious contender, also in this quadrant, is essentially Purdue-lite now.

The most visually striking outlier in Figure 2 is Gonzaga. The Bulldogs are planted in the far middle-left corner of the plane, meaning their their offensive efficiency falls well short of the champion average, as does their defensive efficiency. This is an unusual and historically rare construction for a team with legitimate championship probability, and the model treats it accordingly. Gonzaga's championship case is almost entirely path-dependent: if they avoid high-efficiency offensive teams until late, they can suppress their way to a title. But facing a Purdue or an Arizona or an Michigan in the Sweet Sixteen, Elite Eight, or Final Four would expose exactly their lackluster championship profile. Michigan State, by contrast, is positioned closer to the center of the lower-right quadrant, lacking meaningful efficiency advantages on the offensive end on the floor and a subpar defensive efficiency. Its 2.5% championship probability is powered by path favorability rather than efficiency dominance.

One of the more quietly important observations from this figure is what it says about the relationship between profile quality and championship probability. Duke, on raw efficiency-plane coordinates, is arguably a better "fit" for the historical champion archetype than Michigan — better centered in the upper-right quadrant, its profile exceeding the median champion by a wider margin. And yet Michigan carries a composite championship probability of 23.9% versus Duke's 9.4%. The resolution lies in path structure: Michigan's bracket draw through the Midwest is materially more favorable than Duke's road through the East, and the simulation reflects that difference. The team most likely to win the tournament is not necessarily the team that most closely resembles past champions; it is the team whose profile and path together maximize the probability of surviving six consecutive games against elite competition. In 2026, those two factors are best aligned for Michigan, even if Duke wears the more aesthetically satisfying efficiency costume.

10.3 National Championship Probabilities

Figure 3, while the easiest to interpret in the paper, in some respects is the most deceptive. Legible because the rankings are clear and the bar lengths are unambiguous. Deceptive because the steep drop-off structure encourages a kind of false discretization. There is certainly a temptation to read the field as "Michigan and then everyone else". However, the more accurate characterization is "Michigan clearly, then a four-team race, then a rapidly thinning tail."

Michigan's 23.9% composite probability is not just the highest in the field; it is nearly twice that of the second-ranked team. That kind of separation is unusual. In most tournaments the top two or three teams cluster within a few percentage points of each other. The fact that Michigan has lapped Florida by nearly twelve percentage points reflects not merely a superior efficiency profile, though it has that, but a confluence of profile and path that the model rates as historically uncommon. The Midwest bracket is atypically soft relative to what a one-seed typically faces, and Michigan's two-way efficiency ensures that even if that

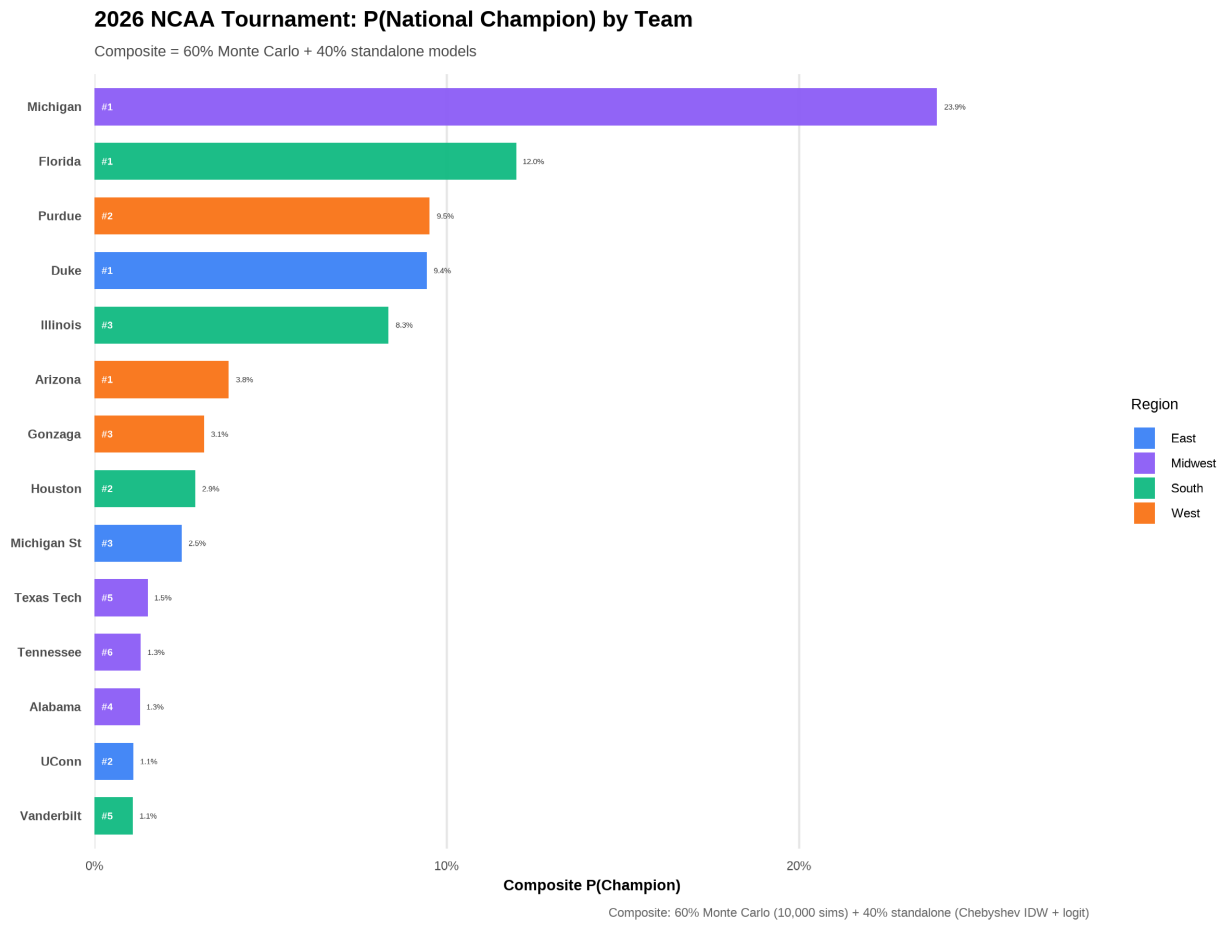

Figure 3: Composite national championship probabilities for all teams with at least a 1% title chance in the 2026 NCAA Tournament. Each bar represents the final composite estimate, constructed as a weighted average of Monte Carlo championship frequency (60%) and standalone team-quality models (40%), normalized to sum to unity across the full field. Bars are color-coded by region.

path stiffens, the team has enough margin to absorb it. As such, the team in the Midwest with the second best odds to cut down the nets is the six-seed Tennessee at approximately 1%.

Florida at 12.0% occupies a clear but lonely second tier. It is far enough ahead of Purdue and Duke (both hovering near 9.4–9.5%) to be considered a distinct profile, but far enough behind Michigan to make a Florida title feel more like a plausible outcome than the model's central expectation. Florida's path through the South is absolutely harder than Michigan's through the Midwest, and the efficiency-plane positioning from Figure 2 already established that Florida's profile is slightly less archetypal. Put those two facts together and the 12.0% figure feels about right: Florida is a serious title contender, perhaps the most complete team in the South, but it needs things to break favorably in a way that Michigan simply does not.

Purdue and Duke are essentially tied at 9.5% and 9.4% respectively, which is an interesting juxtaposition given how different their efficiency profiles are. Purdue is an offensive powerhouse with defensive vulnerabilities; Duke is a more balanced two-way team. Their near-identical composite probabilities reflect the model's implicit weighting of path favorability against profile quality. Purdue's path through the West is difficult with potential second weekend dates of Gonzaga and Arizona, but its offensive efficiency is so extreme that the model gives it substantial credit for grinding through high-quality opposition regardless. Yet, Purdue's path is nowhere close to as challenging as Duke's. Duke's East region could feature potential Round of 32, Sweet Sixteen, and Elite Eight matchups with Ohio St., Kansas, and UConn resulting in its objectively better profile than Purdue being so outweighed by its potential road that Purdue has the better odds to cut down the nets.

Illinois at 8.3% completes what is effectively a four-team cluster from 8–10%, all of whom have realistic but not dominant title cases. Illinois's championship probability is powered primarily by its elite offense — it is, along with Purdue, among the most offensively efficient teams in the field — and by a South bracket that, while containing Florida and Houston, is at least structured in a way that gives Illinois multiple winnable games with a Round of 64 tie against the University of Pennsylvania and a potential Round of 32 matchup against a Caleb Wilson-less, faltering North Carolina squad.

Arizona's drop to 3.8% is the model's most eyebrow-raising statistic. Arizona is a one-seed and the second-ranked team in the KenPom efficiency model. And yet the composite probability model gives it a fraction of what it gives Michigan. The explanation is that Arizona shares the West region with Purdue, Gonzaga, Arkansas, and Wisconsin and its efficiency-plane profile, while above the champion archetype, is not so dominant as to overwhelm that structural disadvantage. It also has a potential sneaky difficult Round of 32 matchup with Utah State. Arizona is a very good basketball team that has been handed a very hard bracket. The 3.8% figure is not an indictment of the team; it is an indictment of the committee's placement decisions.[2]

The tail of the distribution contains a few quiet curiosities. Tennessee at 1.3% as a six-seed is the highest-probability team in the field at its seed line and reflects the same efficiency-plane story told in Figure 1: Tennessee is simply better than a six-seed. Vanderbilt at 1.1% is notable given its seeding as a five; the model gives it approximately the same credit as UConn (also 1.1%), which is a two-seed, against supporting the notion that the East is a monster of a region.

[2]We must not also glance over Arizona's Round of 64 matchup versus Long Island University. Fins up!

10.4 Round-by-Round Survival

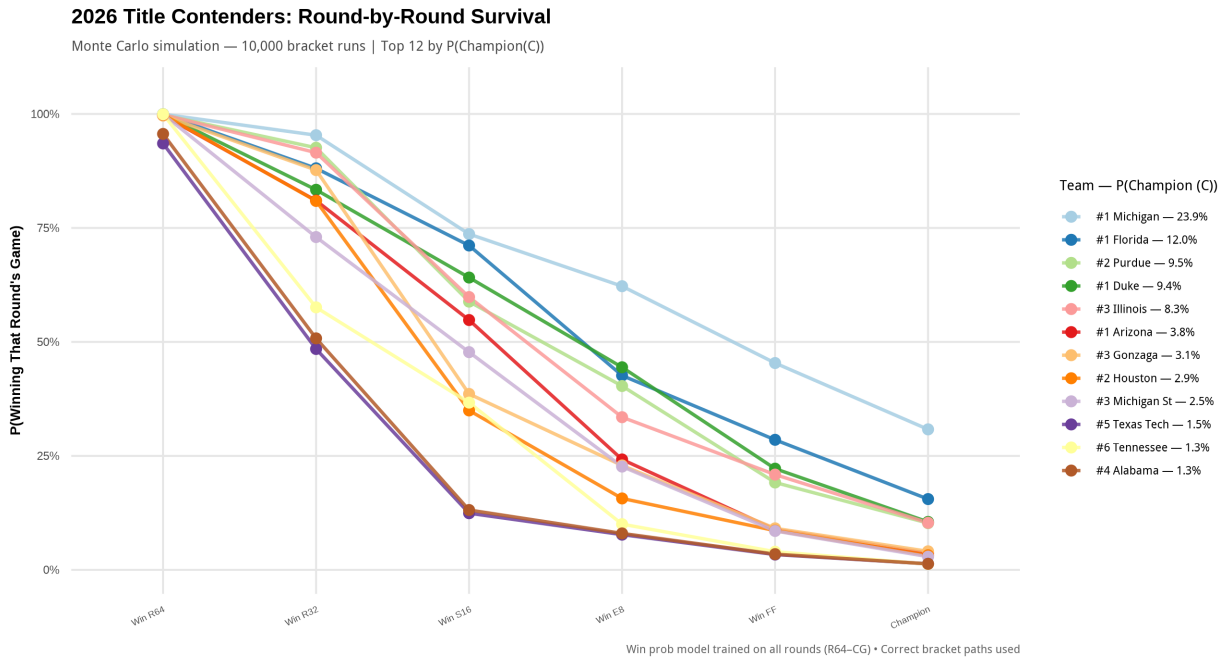

Figure 4: Round-by-round survival probabilities for the leading 2026 title contenders, derived exclusively from the Monte Carlo bracket simulation (10,000 runs). Each curve traces the probability that a team wins the game corresponding to each successive tournament round, from the Round of 64 through the Championship Game. Teams are identified by composite championship probability in the legend. This is not the same probability displayed in the survival chart and is merely used as a legend reference.

Figure 3 is the photograph of the title race to Figure 4's film. It shows not just where teams end up but how they get there, or fail to. The shape of each survival curve is as analytically informative as its endpoint, and several of those shapes have things to say that the final championship numbers alone cannot convey.

All twelve teams begin near or at 100% in the Round of 64. Even the weakest team in this set — Tennessee, a six-seed — clears 75% in the first round, and most of the top contenders are functionally certain to open with a win. The real tournament, in terms of predictive content, begins at the Round of 32, and the divergence that emerges between that node and the Sweet Sixteen is where the model's most important information lives.

Michigan's survival curve is in a category of its own. The light blue line simply refuses to drop at the rate of its competitors. By the Round of 32, Michigan is already above 95% while the field has compressed to the 70–90% range; by the Sweet Sixteen, its probability of winning that game sits around 74%. By the Elite Eight, Michigan is at approximately 62%, roughly fifteen percentage points above its nearest competitor. The Final Four probability of nearly one in two is a remarkable number for a team in a 68-team field and reflects both the bracket structure and the efficiency profile working in near-perfect concert with one another.

Florida's curve is the clearest illustration of the "path dependency" concept introduced elsewhere in this analysis. Florida's early-round win probabilities are high, tracking closely to Michigan's through the Sweet Sixteen. But the South region's competitive density catches up with Florida in Sweet Sixteen and the Elite

Eight, where its survival probability drops more steeply than its underlying efficiency profile alone would predict as the model forecasts meets with 5-seed Vanderbilt and 2-seed Houston. The result is that Florida, despite carrying the second-best championship profile in the field, sees its Final Four probability fall sharply relative to Michigan's.

Purdue and Duke trace remarkably parallel paths through the bracket despite the efficiency-plane differences noted in Figure 2. Both teams' curves begin to diverge meaningfully from the top two after the Sweet Sixteen. Purdue's offensive-powerhouse construction buys it strong early survival probabilities but by the Elite Eight and Final Four the two curves have essentially reconverged.

Illinois presents one of the survival chart's most interesting inflection points. Its Round of 32 win probability is high — north of 90% — which reflects both strong underlying quality and a favorable projected second-round opponent. But the Sweet Sixteen line and subsequent Elite Eight line drop sharply, more so than for either Purdue or Duke. The gap between Illinois's first-weekend strength and its second-weekend fragility is the analytical story of its bracket placement. A potential second weekend for Illinois would include Houston and Florida at the Toyota Center.

Arizona's survival curve is, frankly, the most striking anomaly in the figure. For a one-seed — theoretically among the four best teams in the country — Arizona's curve declines with unusual aggression past the first round. By the Sweet Sixteen it is already tracking below teams seeded three and five in the chart, and by the Elite Eight its survival probability has converged with Gonzaga's. The West is simply a brutal bracket: Purdue at two, Gonzaga at three, Arkansas and Wisconsin in the four and five slots respectively. Arizona's curve is the visual proof of what the broader analysis has been saying: the committee handed a very good team a very hard path, according to my model. What might be most notable is the drop off between the Wildcat's probability of winning the Sweet Sixteen and its probability of winning the Elite Eight, where we see greater than 25 percentage point drop. A potential Elite Eight matchup for Tommy Lloyd's squad would be Purdue.

At the bottom of the chart, Gonzaga, Houston, and Michigan State form a cluster of credible deep-run profiles whose championship probability curves decline sharply after the Sweet Sixteen as well. It is worth taking a closer look at the Tennessee-Alabama-Texas Tech Cluster. The model projects Tennessee to have an easier time with Miami Ohio in the first round than Alabama and Texas Tech will have with Hofstra and Akron respectively. Due to a potential run in to each other in the Round of 32, Tech's and Alabama's curves stay below Tennessee's, who would see a Virginia Cavaliers squad in Philly on Sunday. We must, however, look at the drop between the Round of 32 and Sweet Sixteen. Tennessee sees a drop of approximately 20 percentage points while Alabama and Texas Tech see theirs drop by approximately 35 percentage points as the potential Michigan problem waits in the wings for the latter two in the Sweet Sixteen.

10.5 Round by Round Probabilities of Winning each Round

Table 1 summarizes the top twenty teams by probability of winning the 2026 NCAA Championship, reporting round-by-round win probabilities derived from the Monte Carlo simulation alongside the final composite championship estimate. Teams are ordered by composite championship probability, with KenPom rank, seed, and region provided for contextual reference.

Table 1: Top 20 Teams by Probability of Winning the 2026 NCAA Championship

KP	Team	Seed	Region	Win R64	Win R32	Win S16	Win E8	Win FF	Champ (MC)	Champ (C)
3	Michigan	1	Midwest	100.0%	95.3%	73.7%	62.2%	45.4%	30.8%	23.9%
4	Florida	1	South	100.0%	88.1%	71.2%	42.6%	28.5%	15.5%	12.0%
8	Purdue	2	West	99.8%	92.6%	58.8%	40.3%	19.2%	10.2%	9.5%
1	Duke	1	East	99.9%	83.4%	64.1%	44.4%	22.2%	10.5%	9.4%
7	Illinois	3	South	99.9%	91.5%	59.8%	33.5%	20.9%	10.3%	8.3%
2	Arizona	1	West	99.7%	81.0%	54.8%	24.2%	8.9%	3.6%	3.8%
10	Gonzaga	3	West	99.6%	87.7%	38.6%	22.9%	9.1%	4.1%	3.1%
5	Houston	2	South	99.8%	80.9%	35.0%	15.7%	8.6%	3.2%	2.9%
9	Michigan St.	3	East	99.9%	73.0%	47.8%	22.7%	8.5%	2.9%	2.5%
20	Texas Tech	5	Midwest	93.5%	48.4%	12.4%	7.7%	3.4%	1.3%	1.5%
16	Tennessee	6	Midwest	100.0%	57.6%	36.7%	10.0%	4.0%	1.3%	1.3%
18	Alabama	4	Midwest	95.6%	50.8%	13.1%	8.0%	3.4%	1.3%	1.3%
11	UConn	2	East	99.8%	70.9%	32.6%	12.6%	3.6%	1.0%	1.1%
12	Vanderbilt	5	South	97.9%	66.2%	18.4%	5.4%	2.0%	0.5%	1.1%
17	St. John's	5	East	82.5%	53.4%	18.3%	9.0%	2.6%	0.8%	0.9%
15	Arkansas	4	West	99.9%	55.4%	23.1%	6.8%	1.8%	0.6%	0.8%
6	Iowa State	2	Midwest	100.0%	69.2%	31.2%	6.3%	2.1%	0.5%	0.8%
13	Virginia	3	Midwest	99.5%	42.4%	23.7%	4.6%	1.5%	0.4%	0.7%
22	Wisconsin	5	West	99.9%	44.6%	16.2%	4.2%	0.9%	0.2%	0.6%
19	Louisville	6	East	65.0%	19.8%	9.3%	3.0%	0.6%	0.1%	0.6%

Notes: Win probabilities reflect the chance of winning the specified round game (not merely reaching the round). Champ (MC) denotes Monte Carlo championship probability; Champ (C) is the final composite estimate, normalized to sum to one across the full 64-team field.

The columns labeled Win R64 through Win FF report the probability that a team wins the specified round's game, not merely that it reaches that round. These probabilities are computed directly from simulation frequencies and therefore reflect both underlying team quality and bracket path difficulty. Champ (MC) reports the raw Monte Carlo championship frequency; Champ (C) reports the final composite normalized across the full field. Differences between these two columns highlight the extent to which a team's title equity is driven by path structure versus pure efficiency.

The top of the table is dominated by a narrow set of elite contenders. Michigan stands clearly apart, posting the highest win probabilities at every stage beyond the Round of 32 and a Monte Carlo championship frequency exceeding 30%. Its composite championship probability of 23.9% reflects a combination of elite two-way efficiency and a comparatively favorable projected path through the Midwest region. Florida occupies a clear second position, maintaining strong early-round advancement but exhibiting a sharper decline in Elite Eight and Final Four win probabilities, consistent with a more competitive regional landscape.

Purdue, Duke, and Illinois form a tightly clustered second tier. All three are near locks to advance through the opening weekend, but separation emerges in the later rounds. Purdue's profile is characterized by particularly strong early-round win probabilities followed by declining survival against elite opposition. Duke shows slightly stronger late-round resilience due to a better overall championship profile despite a would be harder path, while Illinois remains competitive through the Sweet Sixteen before encountering increased resistance deeper in the bracket.

A third tier of contenders including Arizona, Gonzaga, Houston, and Michigan State retains meaningful deep-run potential but with substantially lower championship odds. These teams frequently project to reach the second weekend but display rapid probability decay thereafter. Teams such as Texas Tech, Tennessee,

and Alabama exhibit reasonable chances of early advancement but face steep attrition as the tournament progresses. Others including Vanderbilt, St. John's, Wisconsin, and Louisville appear primarily as long-shot outcomes capable of isolated second weekend upsets but lacking the efficiency balance and cumulative path advantages historically associated with final four or national champion teams.

10.6 Cinderella Analysis Results

The Cinderella analysis produces three figures, each capturing a different dimension of Cinderella potential. The first is a composite bar chart that summarizes overall Cinderella likelihood using the weighted measure $\hat{P}_t^{\text{Cinderella}} = 0.25 \cdot \hat{P}_t^{\text{upset}} + 0.75 \cdot \hat{P}_t^{\text{MC,cin}}$. The bar length represents this combined probability, while the annotations report three underlying quantities: the Round of 64 opponent, the upset model's estimated probability of winning that game, and the raw Monte Carlo Cinderella frequency. These annotated values differ because they originate from distinct models: the upset probability is generated by a logistic regression trained only on Round of 64 matchups involving seeds seven and higher, while the Monte Carlo frequency is derived entirely from repeated bracket simulations and therefore reflects path structure rather than isolated matchup characteristics.

10.6.1 Composite Cinderella Probabilities

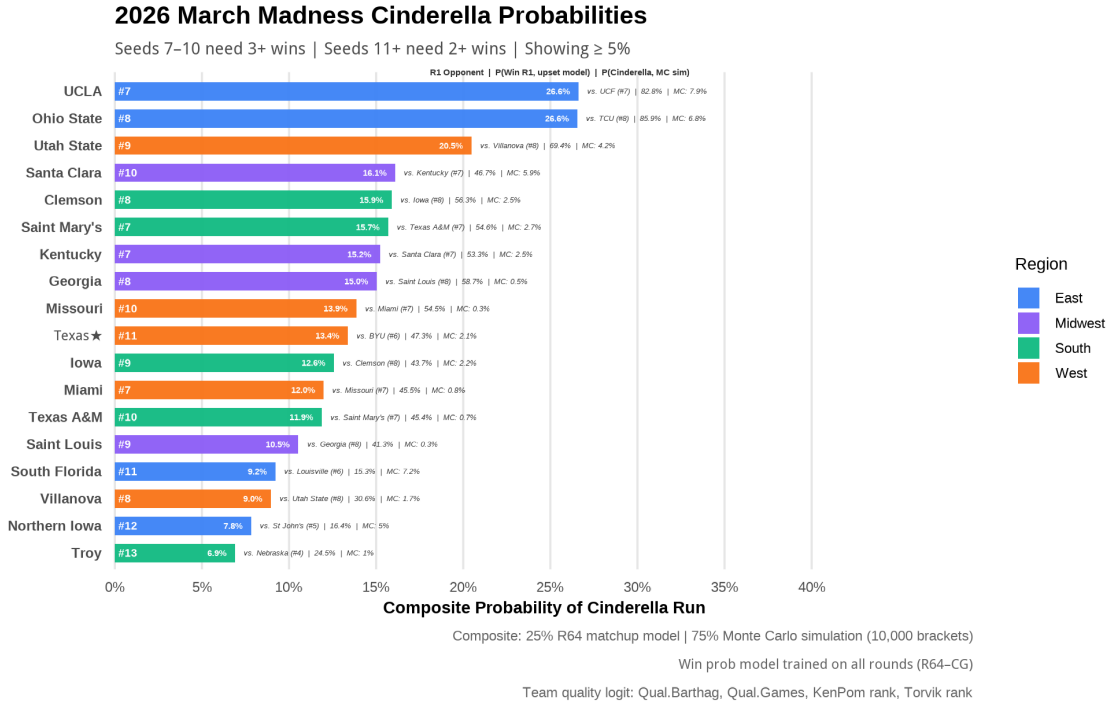

Figure 5: Composite Cinderella probabilities for all teams with at least a 5% Cinderella chance in the 2026 NCAA Tournament. Composite: $0.25 \cdot \hat{P}^{\text{upset}} + 0.75 \cdot \hat{P}^{\text{MC,cin}}$. Annotations report the Round of 64 opponent, specialist upset model win probability, and raw Monte Carlo Cinderella frequency. Bars are color-coded by region.

Figure 5 is the most bracketologically entertaining graph, which is appropriate given its subject matter.

Cinderella stories are the emotional engine of March Madness, and the composite bar chart makes an attempt to identify which teams are most likely to provide them.

At the top of the chart, UCLA and Ohio State are tied at 26.6% — a remarkable coincidence given that they are seeded seventh and eighth respectively in the East region further reinforcing the notion of the East as a gauntlet. UCLA's probability is driven heavily by path favorability: the upset model gives it an 82.8% chance of winning its opening game against UCF, but the Monte Carlo frequency of only 7.9% signals that UCLA's multi-game Cinderella upside is substantially more limited once the bracket deepens. Ohio State tells a nearly identical story: 85.9% win probability against TCU but only 6.8% MC frequency. These are teams almost certain to win their opening games, with substantial but not spectacular chances of extending a run to the Elite Eight, where they would thereafter be classified as Cinderellas.

Utah State at 20.5% is perhaps the field's most interesting true Cinderella candidate. Unlike UCLA and Ohio State, whose composites are dominated by first-round probability, Utah State's case is grounded in a navigable multi-round path. It faces Villanova in Round of 64 at 69.4% win probability and its 4.2% MC frequency, while modest in absolute terms, is among the higher sustained multi-round probabilities in the Cinderella field. A nine-seed reaching the Elite Eight is the archetype of a true Cinderella run, and Utah State is the model's top candidate for that outcome with potential matchups of Arizona in the Round of 32 and Arkansas in the Sweet Sixteen.

Santa Clara at 16.1% as a ten-seed is of note. Its Round of 64 opponent is Kentucky, which the upset model gives it a 46.7% chance of defeating and the 5.9% MC frequency suggests that if Santa Clara wins that game, its subsequent path through the bracket remains survivable with Iowa State in the Round of 32 and the winner of Virginia and Tennessee in the Sweet Sixteen. Santa Clara is, in short, a real Cinderella candidate rather than a chart cameo.

Clemson (15.9%) and Saint Mary's (15.7%) follow closely behind, both representing the archetype of a middling Cinderella candidate: strong enough to win their first game slightly better than half the time but limited in multi-round upside thereby limiting Cinderella capability. Clemson faces Iowa at 56.3% while Saint Mary's faces Texas A&M at 54.6%; neither MC frequency (2.5% and 2.7% respectively) is high enough to suggest sustained deep-run potential.

Texas, marked with a star to denote its selection as a First Four team, appears at 13.4% with a Round of 64 matchup against BYU at 47.3% win probability. The 2.1% MC frequency is modest, but eleven-seed deep runs are well-represented in the historical record, and Texas's quality-adjusted placement in the figure suggests the model views it as a more credible Cinderella than its seed implies. A potential Round of 32 matchup for Sean Miller's Longhorns would be a Braden Huff-less Gonzaga squad looking to get back to the Sweet Sixteen after falling short of doing so a year ago to Houston.

The bottom of the chart features South Florida at 9.2%, Villanova at 9.0%, Northern Iowa at 7.8%, and Troy at 6.9% and represents teams with more limited structural upside. Troy at 6.9% as a thirteen-seed is, as the survival analysis confirms, an almost purely first-round story. Above them lie Miami (FL.), Texas A&M, and Saint Louis whose projections to win their opening ties are close to 50% but whose multi-round upside is significantly limited.

10.6.2 Round-by-Round Cinderella Survival

The second figure in this Cinderella analysis is the round-by-round survival chart. This figure reports $\hat{P}_t(\text{win R64})$, $\hat{P}_t(\text{win R32})$, $\hat{P}_t(\text{win S16})$, and $\hat{P}_t(\text{win E8})$ for each candidate team, calculated exclusively from the Monte Carlo simulation. No standalone team-quality model or first-round upset regression contributes to these values; the survival curves represent the most path-aware view of Cinderella viability and serve as the primary driver of the composite probability.

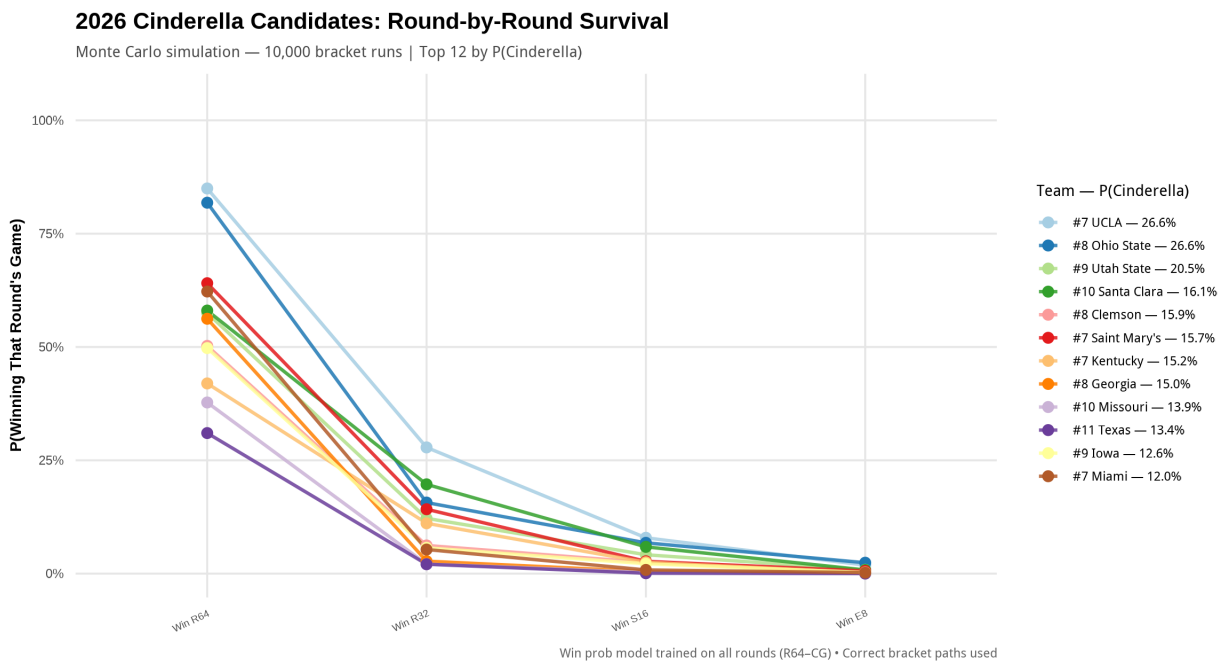

Figure 6: Round-by-round survival probabilities for leading Cinderella candidates in the 2026 NCAA Tournament, estimated purely from the Monte Carlo bracket simulation (10,000 runs). Lines trace the probability of winning each successive round's game from the Round of 64 through the Elite Eight.

Figure 6 is where the Cinderella candidates are sorted into the teams that can actually sustain a run and the teams that are, at best, one-game wonder threats. The distinction between these two profiles is visible in the slope of each line after the Round of 32, and it is steeper for most teams than casual bracketology would suggest.

UCLA's trajectory is the most visually striking in the figure. It begins as the highest of the Round of 64 at near 85% and still holds roughly 29% at the Round of 32 where they would play a Connecticut squad looking to make it past the Round of 32 after their loss last year to eventual national champion Florida. The implication is that UCLA's Cinderella potential is not front-loaded in the way that many lower seeds are: it carries enough underlying quality to make a second-round win actually plausible. After the Round of 32, the curve flattens still, with Sweet Sixteen win probability falling to roughly 10% and Elite Eight probability approaching zero. For the reverse, we can look to Ohio State whose potential second round matchup with heavyweight Duke causes their survival curve to be the steepest of those plotted between the Rounds of 64 and 32. This steepness is so great that 10-seeded Santa Clara actually overtakes Ohio State between the Rounds of 64 and 32, speaking to a difference in quality of opponent between the two teams (Duke vs. Iowa

St.). This also reveals one of the weaknesses of the format of March Madness. Theoretically, 8 and 9 seeds should have an easier time advancing in the tournament, yet Round of 32 matchups with stronger 1-seeds limits their upside compared to the more winnable Round of 32 matchup for 10 seeds of a 2 seed.

Utah State and Santa Clara display some of the survival chart's most encouraging sustained-run profiles among those seeded 9 through 16. Utah State holds meaningful win probability through the Round of 32 and still registers non-trivial Sweet Sixteen survival, reflecting the bracket path that makes it the model's top true-Cinderella candidate. Santa Clara's curve is similar in shape but lower.

Lower down the rankings, while Georgia is projected to win the race to 100 against Saint Louis, their potential Round of 32 tie with Michigan all but eliminates any potential Cinderella upside. Similarly, Texas and Missouri have near zero chances to reach the Sweet Sixteen.

10.6.3 Quality Versus Matchup Context

Figure 7 is the analytical counterpart to Figure 5: where the composite bar chart combines quality and matchup context into a single probability, this scatter plot separates them. The horizontal axis reports $\hat{P}_t^{\text{standalone,cin}}$, a bracket-independent quality signal derived from the Chebyshev model combined with the Cinderella-specific logistic regression. The vertical axis scales this quality score by $\hat{P}_t^{\text{upset}}/\bar{p}_{\text{upset}}$, effectively amplifying or dampening the quality signal depending on how favorable the specific Round of 64 draw is relative to the historical average upset rate. This chart is exploratory and does not feed directly into the composite probability; it serves to distinguish whether a team's Cinderella potential arises primarily from underlying quality or from a favorable opening matchup.

Ohio State is the figure's dominant outlier. Sitting in the far upper-right corner, it combines the highest standalone quality score among all Cinderella candidates with the highest matchup-adjusted probability. It is the rare team for whom both the underlying efficiency profile and the specific Round of 64 draw push in the same direction. Missouri, Santa Clara, Clemson, and Texas also occupy the upper-right quadrant, though more modestly. Iowa's appearance in this quadrant is worth flagging: among the teams above the horizontal mean, Iowa carries the strongest standalone quality signal after Ohio State, meaning its Cinderella potential is less opportunistic and more grounded in actual efficiency-based merit. As noted earlier, if I were to change the operational definition of a Cinderella and instead grouped 10 through 16 seeds together, I would expect Santa Clara to occupy a more isolated upward and rightward position in this figure due to its lower qualifying barriers. Kentucky, Utah State, and UCLA round out the top right quadrant with their more central positions. Of this group, UCLA has the most favorable path and, as such, is positioned the highest vertically. The lack of horizontal differentiation signals the equivalence of this group's metrics.

The upper-left quadrant — weaker quality, favorable draw — is where teams like Saint Mary's and the University of Miami are found. Teams clustered in the lower-left — VCU, South Florida, and several higher-seeded entrants — lack both the efficiency profile and the favorable matchup context historically associated with successful Cinderella runs. Also of note is the University of Northern Iowa and TCU who both have reasonable Cinderella upside based on team quality alone but whose tough first round draws inhibit their ability to advance meaningfully in the tournament.

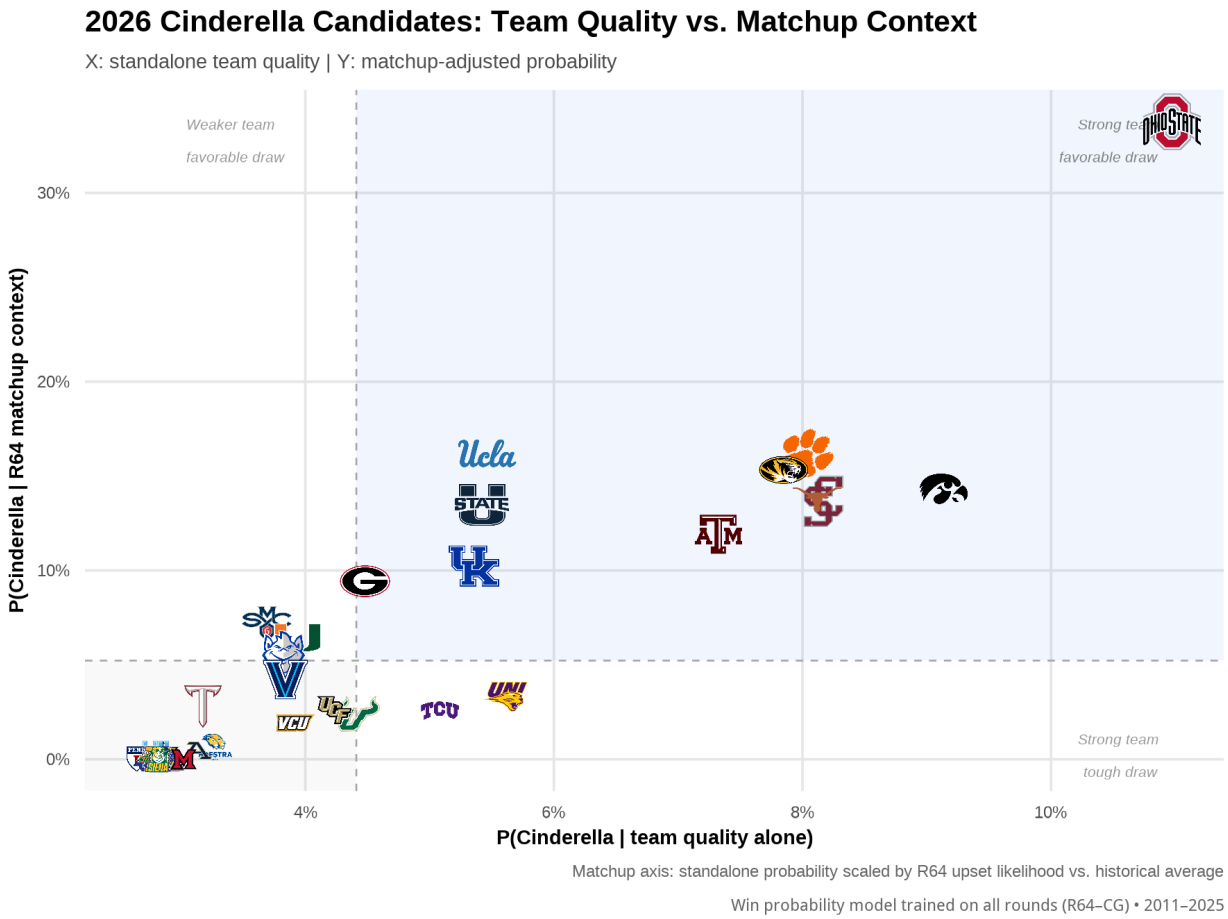

Figure 7: Cinderella candidates decomposed by standalone team quality (horizontal axis) and matchup-adjusted probability (vertical axis), defined as standalone quality scaled by $\hat{P}^{\text{upset}}/\bar{p}_{\text{upset}}$. Dashed lines denote sample means on each axis. Teams in the upper-right quadrant combine strong underlying quality with a favorable projected path.

10.7 Bracket Projections

The three bracket output methods described in the methodology — deterministic, modal, and upset threshold — each reflect a different philosophy about how to translate probability estimates into a single filled-out bracket. They agree more than they disagree, which is itself informative: when independent methods converge on the same outcome, the signal is stronger than any single method alone could claim. Where they diverge, the divergence is analytically meaningful rather than arbitrary, pointing to specific regions of the bracket where path uncertainty is in reality high and where the simulation's variance, rather than its central tendency, becomes the more important quantity.

It is worth pausing before walking through the brackets individually to observe something about the relationship between these three methods and the bracket that any human being, including myself, will actually fill out. The deterministic bracket is what a machine does when it feels no pressure and has nothing to prove. The modal bracket is what 10,000 simulated universes agree on when polled. The upset threshold bracket is what happens when a model is forced to reckon with the fact that the tournament does not care about its central tendency. And the human bracket is something else entirely: it is a document produced under a stew of social pressure, aesthetic preferences, tribal loyalties, and the deeply irrational but emotionally compelling desire to be the person in the office pool who called the Cinderella. Indeed, the human bracket is a confluence of every human irrationality. Furthermore, every bracket analyst who has ever published a model has also quietly wanted to pick an upset that the model did not endorse, just to have something to say if it hits. I, certainly, am not immune to this hedging.

10.7.1 Deterministic Bracket

The deterministic bracket (Figure 8) is the simplest and most conservative of the three outputs. Every game goes to the team the model rates as the more likely winner, full stop. There are no coin flips, no probabilistic decisions, and no accommodation for the structural reality that roughly a third of all tournament games are won by the lower-seeded team. What results is a bracket that would be statistically unusual to observe in practice — a historically typical 64-team field produces 10–15 upsets, and this bracket produces essentially none of consequence past the first round — but that nevertheless serves a useful function: it identifies the model's single most likely path from the opening round to the national championship game, with all variance suppressed.

In the East, Duke advances without interruption, dispatching Siena in the first round before defeating Ohio State, St. John's, and Michigan State to claim the region. The path is clean, reflecting Duke's status as the clear favorite in a region where no other team poses a severe threat to its composite championship probability. Michigan State, seeded third in the East, provides the stiffest theoretical resistance but falls in the Elite Eight, consistent with the survival chart's portrait of a team whose second-weekend probabilities fall meaningfully short of those of the field's elite contenders.

In the South, Florida's march is similarly unimpeded. The Gators handle Prairie View in the first round, then Clemson, Vanderbilt, and Illinois in succession to reach the Final Four. Illinois's Elite Eight loss to Florida is the South region's most consequential result: it confirms the model's view that while Illinois

Figure 8: Deterministic bracket projection for the 2026 NCAA Tournament. Every game is awarded to the team with $W_{ij} > 0.5$ in the precomputed win probability matrix. Michigan cuts down the nets.

carries real title equity, it faces a structurally harder path than its efficiency profile alone would suggest. Houston, seeded second in the South, gets past the Big West's Idaho in the Round of 64 and the West Coast Conference's Saint Mary's in the Round of 32 but does not have enough to get past the Fighting Illini in the first game of the second weekend.

In the West, Purdue advances to the Final Four by defeating Queens, Miami, Gonzaga, and Arizona in sequence. Arizona's Elite Eight loss to Purdue is the West's defining result and the bracket's clearest expression of the committee's placement predicament: the model's top-two efficiency teams meet in the Elite Eight, eliminating a genuinely elite one-seed before the Final Four. Gonzaga reaches the Sweet Sixteen but falls to Purdue, consistent with the survival chart's portrait of a team whose offensive limitations are exposed when matched against an elite offensive opponent.

In the Midwest, the Michigan Wolverines defeat Howard, Georgia, Texas Tech, and Tennessee on the way to the Final Four. Tennessee's Elite Eight exit is the Midwest's headline result — the six-seed had been flagged throughout this analysis as badly underseeded, and the deterministic bracket confirms that the model expects it to reach the Elite Eight, a remarkable projected performance for a six-seed that validates the seed-versus-projected-wins analysis from Figure 1.

The championship game pits Michigan against Florida, with Michigan winning. This is the model's modal championship game and its most probable individual outcome: the two teams with the highest composite championship probabilities, meeting for the title, with the team carrying the more favorable path and the stronger two-way efficiency profile prevailing. The deterministic bracket's value is not that it will happen this way — it almost certainly will not, because March does not operate deterministically — but that it accurately represents the model's single best guess about how the tournament unfolds when variance is removed from the equation.

10.7.2 Modal Bracket

Figure 9: Modal bracket projection for the 2026 NCAA Tournament, derived from 10,000 Monte Carlo simulations. Each game is awarded to the team that most frequently occupied that bracket slot across all simulations. Michigan cuts down the nets.

The modal bracket (Figure 9), derived from 10,000 simulated tournaments, is the most methodologically principled of the three outputs. Rather than advancing the team the model rates as individually more likely to win each game, it advances the team that most frequently appeared in each bracket slot across the full simulation distribution. The distinction matters: a team can be the modal occupant of a Final Four slot not because it is the most likely winner of any single game, but because the compounding of its win

probabilities across multiple rounds makes it the most frequently surviving team at that stage.

The first thing worth noting about the modal bracket is how little it differs from the deterministic one. Both project Duke out of the East, Florida out of the South, Purdue out of the West, and Michigan out of the Midwest. Both produce a Michigan–Florida championship game won by Michigan. The paths through each region are nearly identical at every stage that matters. This is, in one sense, unsurprising: when a team is actually the most likely winner of a game, it tends to also be the most frequent winner across thousands of simulations, and the two methods should agree wherever the model's probability estimates are unambiguous. But the degree of agreement here is still striking. It means the model is not hedging. It is not producing a distribution of plausible outcomes where no single path dominates — it is producing a distribution where one path dominates so thoroughly that 10,000 simulated universes, polled independently, return essentially the same bracket that a simple majority-wins rule produces. The modal model and a deterministic model tell the same story, and as such the story is probably true.[3] The noise has been averaged out and what remains is signal, and the signal says Michigan. However, there is only one matchup difference between the two brackets. The modal bracket has Texas Tech beating Alabama in the Round of 32 whereas the deterministic bracket has the reverse.

10.7.3 Upset Threshold Bracket

The upset threshold bracket (Figure 10), with $p_{\min} = 0.25$ and random seed fixed at 42, is the method most willing to surface plausible non-consensus outcomes and as such it delivers a Final Four that is different from the other two outputs while ultimately arriving at the same champion. Any game where the underdog carries at least a 25% chance of winning is decided by a probabilistically weighted coin flip rather than a deterministic favorite-wins rule. Should I lower or increase the threshold probability, we would expect more or less upsets respectively. Blowout mismatches still go to the favorite while competitive games get flipped. The result is a bracket that reflects the uncertainty embedded in the model's probability estimates and this time the variance surfaces in the East and the South rather than culminating in an alternate champion.

In the East, the upset threshold bracket's most consequential flip sends UConn advancing through the bracket and ultimately out of the region to the Final Four — a result that eliminates Duke, the chalk favorite of the region, before the national semifinals. This is the bracket's single most dramatic departure from the other two outputs. Duke had been the consensus East representative in both the deterministic and modal methods, and UConn's Final Four appearance here is not an artifact of noise but a reflection of a probability: the Huskies' win probability against Duke in an Elite Eight setting clears the 25% threshold, and the weighted coin landed their way. A UConn return to the Final Four would certainly be a storyline, and the upset threshold bracket is doing exactly what it is designed to do by surfacing it. Another consequential flip is that of Pat Kelsey's Louisville squad over Michigan State in the Round of 32. Again, the Cardinal's win probability against Sparty clears the 25% threshold, and the weighted coin similarly landed their way.

In the South, the bracket sends Houston through the bottom half, while Florida claims the top half, consistent with other methods. The South's upset threshold divergence from the deterministic bracket is primarily in the following outcomes: Texas A&M advances over Saint Mary's in the second round before Houston

[3] It most certainly will not be.

Figure 10: Upset threshold bracket for the 2026 NCAA Tournament ($p_{\min} = 0.25$, random seed = 42). Games where the underdog's win probability meets or exceeds 25% are decided by a weighted coin flip using model probabilities; all others go deterministically to the favorite. Michigan cuts down the nets.

dispatches them; and VCU beats North Carolina in the first round before falling to Illinois which are both results the Cinderella analysis flagged. Florida reaches the Final Four out of the South in this bracket, displacing Houston, which represents the upset threshold method's most significant South-region departure: Illinois, the consensus winner in the deterministic and modal outputs, falls to Houston in the Sweet Sixteen when the weighted coin resolves a competitive matchup in the Cougar's favor.

In the West, the path plays out different to the other methods. Purdue advances through a Missouri squad that defeated Miami (FL.), Gonzaga, and Arkansas notably on to the Final Four. BYU falls to Sean Miller's Texas squad where they thereafter lose to Gonzaga, a flip that reshuffles the bracket's geometry without ultimately changing the West's representative. In the top half of the region, the result to note is an Arizona loss to Darius Acuff and Arkansas. Again we have a competitive matchup that the weighted coin resolved. The coin though, fails to help Arkansas against Matt Painter's Purdue as mentioned already. In this West region, we see perhaps some of the greatest effects of the upset threshold, seeing multiple teams making it farther than the deterministic and modal brackets both predicted.

In the Midwest, Michigan's path is largely intact, with the notable exception of Santa Clara advancing over

Iowa State in the Round of 32, another credible mid-major upset that the Cinderella analysis identified before falling to Virginia and Iowa State's absence from the second weekend goes unremarked by the model, which had Michigan winning the Midwest regardless of the bracket's lower half composition. Virginia also makes use of the coin in the Round of 32, defeating Tennessee.

The championship game is Michigan over Florida. This is perhaps the upset threshold bracket's most instructive output: even a method specifically designed to surface variance and realize plausible non-consensus outcomes cannot dislodge Michigan nor Florida from the championship. When the weighted coin lands against Duke in the East, Arizona in the West, and Illinois in the South, the bracket still gives the nod to the Wolverines. That is a strong signal. Michigan's 23.9% championship probability is not just the model's central tendency — it is robust to the kinds of probabilistic shocks the upset threshold method introduces. The bracket confirms this.

10.7.4 My Bracket

The deterministic bracket is emotionally unsatisfying. It picks the favorite in every game, produces a clean chalk bracket that nobody in any office pool has ever won and reads like a bracket filled out by someone who has never watched college basketball and is simply following instructions and applying basic integer ordering. The modal bracket converges on the same Michigan–Florida championship game and is largely indistinguishable from the deterministic output. The upset threshold bracket, by contrast, gave us UConn knocking off Duke and plausible Arkansas and Virginia Elite Eight runs among other results (e.g. Louisville def. Michigan State). Such results are either the model being honest about variance or random seeds doing what random seeds do, depending on how charitable one is feeling.

My bracket (Figure 11) is somewhere in the middle as it accepts the model's conclusion that Michigan will cut down the nets come April but succumbs to the upsets and story lines that every human bracket does. The bracket that results is not a repudiation of the model; it is the model filtered through a set of aesthetic and strategic preferences that no logistic regression can fully encode.Indeed, the human mind is a mysterious thing.

In the East, I largely reject the model. St. John's advances out of the region with wins over Northern Iowa, Kansas, Duke, and UCLA behind hall of fame coaching from Rick Pitino and All-American level play from Zuby Ejiofor despite their lackluster non-conference performance. I have TCU defeating Ohio State in the first round largely because TCU was so impressive in Big 12 play and Michigan State advancing to the Sweet Sixteen before falling to UCLA — a repeat of the Big 10 Tournament. I have UCLA advancing over UConn to the Sweet Sixteen, another of my bracket's meaningful departures from the model's central tendencies and a direct rejection of my analyses most consistent findings. It is an upset call grounded in UCLA's quality signal from Figure 7 rather than pure wishful thinking, though I will admit there is also something aesthetically appealing about Mick Cronin's Bruins behind Donavan Dent making a deep run. St John's then ends UCLA's tournament in the Elite Eight.[4] I am not drinking the Bryan Hodgson and South Florida Kool-Aid either.

In the South, I make more departures from the deterministic and modal outputs: Houston advances to the

[4]Should this not happen and UCLA bow out earlier, watch out Liberty Bell.

Figure 11: My 2026 NCAA Tournament bracket. Michigan over Houston.

Final Four rather than Florida or Illinois. I have Florida defeating Iowa and then losing to Tyler Tanner and Vanderbilt whereupon Houston eliminates Vanderbilt in what would be a historically electric game: a Vanderbilt team coming off of wins over Nebraska and Florida falling to Kelvin Sampson's program in a game that, if played in Houston, would be one of the loudest environments in tournament history thereby sending the Cougars, once again, to Indianapolis. I am taking into account both the home-adjacent advantage and the fact that I, genuinely, want Kelvin Sampson to succeed. The model would not endorse this sequence. The model is probably right. I also have VCU beating North Carolina in the first round — VCU's lower-left quadrant placement in Figure 7 makes it among the least analytically supported upsets in the field, and I picked it anyway. This is emblematic of the nature of bracket-filling: the model tells you not to pick VCU and some part of your brain remembers Shaka Smart's 2011 squad and picks it anyway.

In the West, I align fully with the upset model more so than any other model: Purdue to the Final Four via Arkansas in the Elite Eight. Arkansas advances out of the Wisconsin–Arkansas second-round matchup, which sets the table for the West's most intriguing regional narrative: the Arizona–Arkansas Sweet Sixteen. I believe that Darius Acuff is right up there as one of the best players in the West — Sean Miller would surely agree — and as such I think he can drag this Arkansas squad past Tommy Lloyd's Arizona, further extending Lloyd's remarkable streak of not advancing past the Sweet Sixteen. Purdue then defeats Arkansas in the Elite Eight, with Purdue's offensive firepower ultimately the decisive factor. The Boilermakers advance to the Final Four as the West's consensus representative across all four brackets. I have to admit, there seems to be some magical carpet ride brewing for Purdue behind Braden Smith and the rest of the seniors on that squad and I have certainly been swept up in the feelings around them the past couple of weeks.

In the Midwest, I follow the model with respect to Michigan and a couple of other matchups, e.g., Georgia beats Saint Louis in the race to 100, but there are some significant deviations. First, I have John Groce's Akron Zips defeating Texas Tech in the Round of 64 and ex-NBA-er Speedy Claxon's Hofstra over Alabama in a guard heavy matchup, results that would be some of the first round's most surprising outcomes. Second, and more dramatically, I have Virginia advancing to the Elite Eight rather than Tennessee. The deterministic and modal outputs have Tennessee in the Midwest regional final, and the seed-versus-projected-wins analysis flags the Vols as perhaps the most underseeded team in the field. I am, on some level, simply tired of agreeing with the model and want another Midwest pick that reflects independent judgment. On another level, I go to Virginia, and want the chance to enjoy more of Ryan Odom's criminally underrated squad. Give me Sam Lewis, Chance Mallory, and Thijs De Ridder getting past a Santa Clara Broncos team coming off an upset over the fighting TJ Otzelbergers in the Round of 32 (a result I have embraced enthusiastically) before advancing to an Elite Eight date with Michigan. Michigan then defeats Virginia in the Elite Eight, consistent with all model outputs. Yaxel Lendeborg and company are simply too much for the Cavaliers and the Wolverines advance to the Final Four as the prohibitive Midwest favorite they have always been.

The championship game is Michigan over Houston. The deterministic and modal brackets both project Michigan–Florida but a tournament in which Houston, Kingston Flemmings, and Kelvin Sampson reach the Final Four is the one I want to watch, and I think that Houston's defensive construction as always is dangerous enough to beat anyone not named Michigan. Michigan is the pick because the model says Michigan is the pick, and at the end of a long analysis, I think I should just trust the model. Everything else in the bracket is noise layered on top of a signal that was always pointing toward Ann Arbor.

11 Limitations and Injury Adjustments

Several limitations should be noted when interpreting the results of this analysis. First, all input features are based on season-long aggregate metrics. As a result, the model cannot account for late-season changes in team performance such as momentum, recent form, or injuries that occur after the regular season concludes. These dynamics often influence tournament outcomes but are not reflected in season-average statistics.

Second, the historical sample used to estimate the deep-run models is relatively small. The dataset includes only 14 NCAA tournaments from 2011 through 2025 (excluding the canceled 2020 tournament), yielding just 56 Final Four observations and 14 championship observations. Such small samples introduce estimation noise and limit the precision with which the probabilities of the deepest tournament outcomes can be estimated.

Third, while all NCAA tournament games are played at nominally neutral sites, the model does not incorporate any adjustment for geographic proximity or crowd effects. In practice, teams that play closer to their home region often benefit from stronger fan support, which can create a de facto home-court advantage that is not captured in the present framework. For instance, this year Houston, should they advance to the second weekend, would play their games at the Toyota Center in Downtown Houston, allowing Cougars fans to show up in droves. Also, Missouri will play its Round of 64 games 100 miles from their campus in St. Louis while their opponent, the higher seeded Miami (FL.) will have to travel substantially farther.

Fourth, although the Monte Carlo simulation correctly enforces bracket paths, the underlying game probabilities treat each matchup as independent. Factors such as accumulated fatigue and opponent-specific preparation across consecutive games are therefore not explicitly modeled.

Fifth, the win probability model is estimated using a single pooled specification across all six rounds of the tournament. While this approach improves statistical power, it implicitly assumes that the influence of variables such as $\Delta\texttt{WAB}$ or efficiency margins is constant across rounds. In reality, the determinants of early-round outcomes may differ from those governing late-round matchups, suggesting that round-specific coefficients would be a natural extension.

Sixth, among the mid-major teams in particular, there is a growing trend for power conference schools to avoid scheduling certain stronger mid-majors as they have no reason to do so. For example, Miami Ohio, High Point, and Akron had a hard time finding power conference teams to schedule them, resulting in their predictive metrics not being as high as their power conference counterparts due to a lackluster non-conference strength of schedule and the typical strength of schedule of a mid-major conference slate. Thus, these teams lose meaningful opportunities to move up in certain metrics, leading to the model underestimating such teams.

Finally, the model does not directly incorporate player-level injury information or off-the-court developments. For example, recent events involving Aden Holloway of Alabama, along with injuries to players such as L.J. Cason of the University of Michigan, Caleb Wilson of the University of North Carolina, Richie Saunders of BYU, JT Toppin of Texas Tech, Mikel Brown Jr. of Louisville, and Caleb Foster and Patrick Ngongba II of Duke among others introduce uncertainty that team-level statistics cannot easily capture.

While injury-adjusted metrics exist, notably those produced by Evan Miyakawa, they are not publicly downloadable without subscription access. Consequently, without detailed player-level data, it is difficult to estimate how such developments may affect the projected performance of these teams.

12 Conclusion

This paper has developed and applied a multi-component forecasting framework for the 2026 NCAA Tournament, combining matchup-level logistic regression, standalone team-quality models, Chebyshev distance-weighted nearest-neighbor classification, and Monte Carlo bracket simulation into a set of composite probability estimates that are simultaneously grounded in team efficiency and sensitive to bracket path structure.

Michigan enters the tournament as the model's prohibitive favorite, with a composite championship probability of 23.9% which is nearly double that of second-ranked Florida. That number is not a product of any single model component; it is robust across the logistic regression, the nearest-neighbor classifier, and 10,000 simulated tournaments, and it survives the probabilistic shocks introduced by the upset threshold bracket. Behind Michigan, the analysis reveals a four-team championship race among Florida, Purdue, Duke, and Illinois, each carrying between 8 and 12% composite probability and each representing a structurally distinct title profile. Florida is the most complete team in the South and the model's clearest second-best, but its more competitive path prevents it from approaching Michigan's probability. Purdue is an offensive powerhouse whose defensive vulnerabilities create specific late-round exposure. Duke carries a well-rounded efficiency profile undermined by a brutal East region draw. Illinois is perhaps the field's most path-sensitive contender: elite offensively, dangerous through the first weekend, but facing a second-weekend landscape that could feature Houston and Florida in back-to-back games at the Toyota Center.

The Cinderella analysis identifies UCLA, Ohio State, and Utah State as the 2026 field's most credible higher-seed deep-run candidates. Santa Clara, Texas, and Missouri are among those seeded 10-16 that also feature on the list of teams with both the quality and the matchup context to produce some memorable March moments.

Several limitations temper these conclusions. The historical sample underlying the deep-run models spans only fourteen tournaments, producing small observation counts at the Final Four and championship level that introduce estimation noise. The win probability model assumes constant coefficients across all six rounds, an assumption that likely understates round-specific dynamics in the later stages of the tournament. Injuries and late-season form which are factors that frequently determine outcomes in single-elimination play are not captured by season-aggregate efficiency metrics, and several teams in the 2026 field have been affected by player availability issues that the model cannot fully reflect.

What the framework can claim is that it provides a principled and internally consistent translation of available team-quality information into probability estimates that respect both the signal embedded in efficiency metrics and the structural uncertainty imposed by single-elimination bracket play. The perfect bracket remains as elusive as ever. My model will not be right, but to the extent that I can, I hope the nets

in Indianapolis will be cut down by the Maize and Blue come early April.[5] Happy March everyone and Wahoowa!

A Appendix: Historical Profiles by Round Reached

Table 2 reports the average team-quality profile by tournament round reached, estimated over 2011–2025. Teams that advance further are systematically better across every dimension, validating the use of these metrics as predictors.

Table 2: Average Barttorvik/KenPom metrics by deepest round reached, 2011–2025

Round	n	barthag	adjoe	adjde	WAB	KP Rank
R64 exit	448	0.729	109.7	99.3	−0.6	79.2
R32 exit	223	0.856	113.5	96.1	2.8	36.8
Sweet 16	113	0.904	116.1	94.5	4.6	21.9
Elite 8	56	0.920	117.4	93.9	6.0	17.0
Final Four	28	0.921	117.9	93.9	5.6	20.0
Runner-Up	14	0.944	119.5	92.1	7.1	11.3
Champion	14	0.962	122.6	91.8	9.0	5.6

[5] I would be more than happy to be wrong about this if it meant the 'Hoos were going to Indy.